NARAYANA
IIT ACADEMY
INDIA

40+ YEARS OF EXCELLENCE

Sec: SR.IIT_*CO-SC(MODEL-A,B&C) Date:28-12-23
Time: 3HRS Max. Marks: 300

Name of the Student: ____________________ H.T. NO:

28-12-23_SR.STAR CO-SUPER CHAINA(MODEL-A,B&C)_JEE-MAIN_GTM-3_SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

MATHEMATICS: TOTAL SYLLABUS

SUBJECT	MISTAKES		
	JEE SYLLABUS Q'S	JEE EXTRA SYLLABUS Q'S	TOTAL Q'S
MATHS			
PHYISCS			
CHEMISTRY			

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

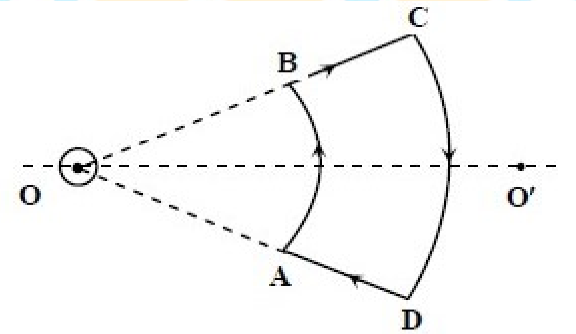
1. Two coherent sources of light interfere. The intensity ratio of two sources is $1:4$. For this interference pattern if the value of $\frac{I_{max}+I_{min}}{I_{max}-I_{min}}$ is equal to $\frac{2\alpha+1}{\beta+3}$, then $\frac{\alpha}{\beta}$ will be

A) 1.5 B) 2 C) 0.5 D) 1

2. Two satellites A and B, having masses in the ratio 4 : 3, are revolving in circular orbits of radii 3r and 4r respectively around the earth. The ratio of total mechanical energy of A to B is.

A) 9 : 16 B) 16 : 9 C) 1 : 1 D) 4 : 3

3. An infinite current carrying uniform wire passes through point O and in perpendicular to the plane containing a current carrying loop ABCD as shown in the figure. Choose the correct option(s)



A) Net force on the loop is zero

B) Net torque on the loop is zero

C) As seen form 'O'', the center of mass of the loop moves towards 'O'.

D) As seen from 'O'', the center of mass of the loop moves away from 'O'.

4. The amplitude of electric field at a distance r from a point source of light of power P is (taking 100% efficiency)

A) $\sqrt{\frac{P}{2\pi r^2 c\varepsilon_0}}$ B) $\sqrt{\frac{P}{4\pi r^2 c\varepsilon_0}}$ C) $\sqrt{\frac{P}{8\pi r^2 c\varepsilon_0}}$ D) $\frac{P}{2\pi r^2 c\varepsilon_0}$

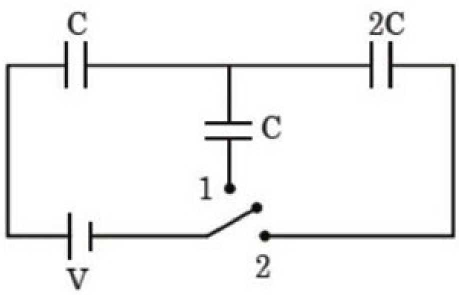
5. An iron rod of length L and magnetic moment M is bent in the form of a semicircle. Now its magnetic moment will be

A) M B) $\frac{2M}{\pi}$ C) $\frac{M}{\pi}$ D) $M\pi$

6. In a football game, a player wants to hit a football from the ground to one of his teammates, who is running on the field. Take hitter position as origin and receiver's initial positions as $2\hat{i}+3\hat{j}$, where $\hat{i}$ and $\hat{j}$ are in the plane of field. Football's initial velocity vector is $2\hat{i}+5\hat{j}+25\hat{k}$ and in the subsequent run receiver displacement is $5\hat{i}$ and $8\hat{j}$, then $2\hat{i}+4\hat{j}$ and then $6\hat{j}$. How far is the receiver from the football when football lands on ground? (assume $\vec{g}=-10\hat{k}$)

A) $\sqrt{10}$ B) $\sqrt{17}$ C) $\sqrt{26}$ D) $\sqrt{13}$

7. In the given circuit diagram, switch was connected to position 1 for long time. At t = 0, switch is shifted from position 1 to position 2. Find the final charge on capacitor 2C.



A) $\frac{CV}{6}$ B) $\frac{CV}{3}$ C) $\frac{2CV}{3}$ D) $\frac{4CV}{3}$

8. **STATEMENT – 1:** Two teams having a tug of war always pull equally hard on one another. (Ignore mass of rope)

STATEMENT – 2: The team that pushes harder against the ground, in a tug of war, wins.

A) Statement – 1 is True, Statement – 2 is True; Statement – 2 **is** a correct explanation for Statement – 1.

B) Statement – 1 is True, Statement – 2 is True; Statement – 2 **is NOT** a correct explanation for Statement – 1.

C) Statement – 1 is True, Statement – 2 is False.

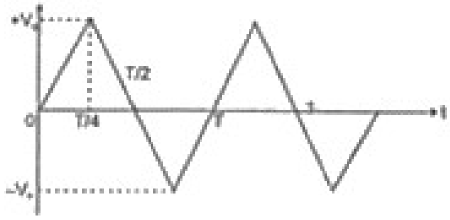
D)Statement – 1 is False, Statement – 2 is True

9. A bag is gently dropped on a conveyor belt moving at a speed of 2 m/s. The coefficient of friction between the conveyor belt and bag is 0.4. Initially, the bag slips on the belt before it stops due to friction. The distance travelled by the bag on the belt during slipping motion, is :

[Take g = $10\,m/s^{-2}$]

A) 2 m B) 0.5 m C) 3.2 m D) 0.8 ms

10. The voltage time (V – t) graph for triangular wave having peak value V_0 is as shown in figure.



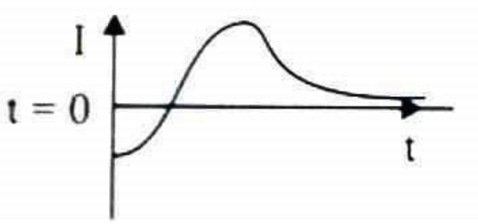
The rms value of V in time interval from t = 0 to $\frac{T}{4}$ is :

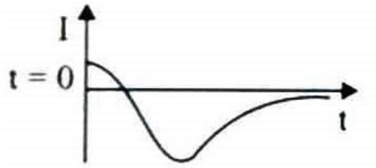
A) $\frac{V_0}{\sqrt{3}}$ B) $\frac{V_0}{2}$ C) $\frac{V_0}{\sqrt{2}}$ D) $\frac{2V_0}{\pi}$

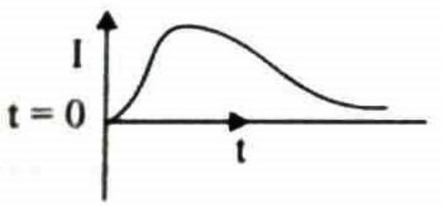
11. A hollow spherical shell at outer radius R floats just submerged under the water surface. The inner radius of the shell is r. If the specific gravity of the shell material is $\frac{27}{8}$ w.r.t water, the value of r is : (Given $\sqrt[3]{19} = \frac{8}{3}$)

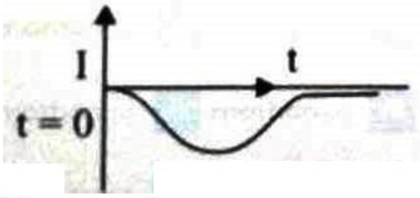
A) $\frac{8}{9}$R B) $\frac{4}{9}$R C) $\frac{2}{3}$R D) $\frac{1}{3}$R

12. A very long solenoid of radius R is carrying current I(t) = Kte^{-at} $(k>0)$, as a function of time $(t \geq 0)$. Counter clockwise current is taken to be positive. A circular conducting coil of radius 2R is placed in the equatorial plane of the solenoid and concentric with the solenoid. The current induced in the outer coil is correctly depicted,

A) 

B) 

C) 

D) 

13. For a nucleus ${}_{Z}^{A}X$ having mass number A and atomic number Z

A. The surface energy per nucleon $(b_s) = a_1 A^{2/3}$

B. The coulomb contribution to the binding energy $b_c = -a_2 \dfrac{Z(Z-1)}{A^{4/3}}$

C. The volume energy $b_v = a_3 A$

D. Decrease in the binding energy is proportional surface area.

E. While estimating the surface energy, it is assumed that each nucleon interacts with 12 nucleons,

(a_1 , a_2 and a_3 are constants)

Choose the most appropriate answer form the options given below;

A) C , D only B) B , C, E only C) A , B , C , D only D) B , C only

14. In a Vernier calipers having 10 VSD, the Vernier constant is 0.1 mm. when the jaws are closed, zero of Vernier lies to the left of zero of main and 7^{th} VSD coincides with a main scale division. When a cylinder is placed between the jaws the main scale reading was 7.7 cm and Vernier scale read 8 divisions. What is the diameter of the cylinder?

A) 78.1 mm B) 77.5 mm C) 77.8 mm D) 78.5 mm

15. Two cylindrical vessels of equal cross – sectional area 16 cm^2 contain water upto heights 100 cm and 150 cm respectively. The vessels are interconnected so that the water levels in them become equal. The work done by the force of gravity during the process, is [Take , density of water = 10^3 kg/m^3 and g = 10 ms^{-2}]

A) 0.25 J B) 1 J C) 8 J D) 12 J

16. In the expression for V_{rms} of a mixture of 2 gases at same temperature, $V_{rms}=\sqrt{\frac{3RT}{M}}$, the molecular weight M is

A) Arithmetic mean of molecular weights

B) Harmonic mean of molecular weights

C) Geometric mean of molecular weights

D) Larger Molecular weight of the two

17. A particle executing SHM is given by equation $x=A\cos(\omega t+\phi)$. When particle is at half the amplitude with positive velocity initially, the initial phase ϕ is

A) $\frac{\pi}{6}$ B) $\frac{\pi}{3}$ C) $\frac{5\pi}{3}$ D) $\frac{5\pi}{6}$

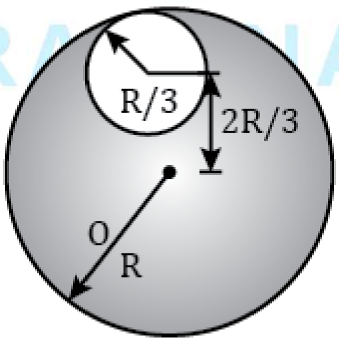
18. A monatomic ideal gas sample is given heat Q. One half of this heat is used as work done by the gas and rest is used for increasing its internal energy. The equation of process in terms of volume and temperature is

A) $\frac{V^2}{T^3}$ = constant B) $\frac{V^2}{\sqrt{T}}$ = constant C) VT^3 = constant D) $V^2\sqrt{T}$ = constant

19. Five point charges each + q , are placed on five vertices of a regular hexagon of side L. The magnitude of the force on a point charge of value – q placed at the centre of the hexagon (in newton) is

A) Zero B) $\frac{\sqrt{3}q^2}{4\pi\epsilon_0 L^2}$ C) $\frac{q^2}{4\pi\epsilon_0 L^2}$ D) $\frac{q^2}{4\sqrt{3}\pi\epsilon_0 L^2}$

20. A disc has mass 9 m and radius R. A hole of radius R/3 is cut from it as shown in the figure. The moment of inertia of remaining part about an axis passing through the centre 'O' of the disc and perpendicular to the plane of the disc is :



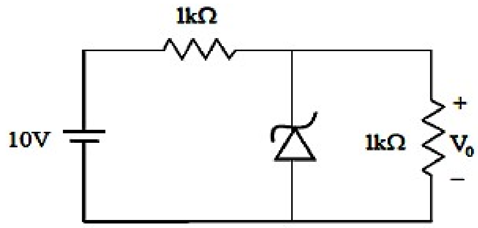
A) $8mR^2$ B) $4mR^2$ C) $\frac{40}{9}mR^2$ D) $\frac{37}{9}mR^2$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

21. In the circuit shown below, the Zener diode is ideal and the zener voltage is 6V. The output voltage V_0 (in volts) is ________________

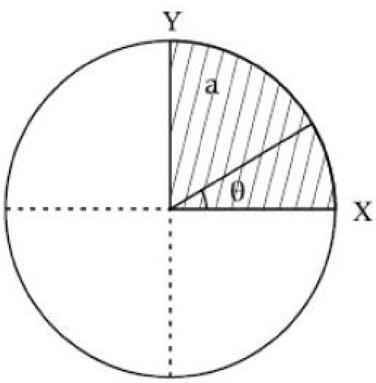


22. The energy (in eV) that should be added to an electron, to reduce its de – Broglie wavelength from 1×10^{-9} m to 0.5×10^{-9} m will be x times the initial energy.

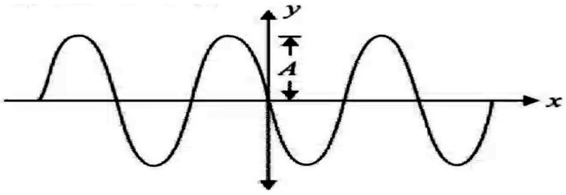
23. The X – Y plane be taken as the boundary between two transparent media M_1 and M_2. M_1 in $Z \geq 0$ has a refractive index of $\sqrt{2}$ and M_2 with $Z < 0$ has a refractive index of $\sqrt{3}$. A ray of light travelling in M_1 along the direction given by the vector $\vec{A} = 4\sqrt{3}\hat{i} - 3\sqrt{3}\hat{j} - 5\hat{k}$, is incident on the plane of separation. The value of difference between the angle of incident in M_1 and the angle of refraction in M_2 will be ___________ degree.

24. A body of mass 500g moves along x axis such that its velocity varies with displacement x according to relation $v = 10\sqrt{x}$ m/s. Force acting on the body is ______________ N

25. The disc of mass M with uniform surface mass density is shown in the figure. The centre of mass of the quarter disc (the shaded area) is at the position $\frac{x}{3}\frac{a}{\pi}, \frac{x}{3}\frac{a}{\pi}$ where x is . (Round off to the Nearest integer) [a is the radius of disc as shown in the figure]



26. A progressive wave travelling along the positive x - direction is represented by $y(x,t)$ = A sin $(kx-\omega t+\phi)$. Its snapshot at t = 0 is given in the figure.



For this wave, if the phase ϕ is $\frac{x\pi}{2}$ then x is ___________. $(\phi \in [0, 2\pi])$

27. A circular disc reaches from top to bottom of an inclined plane of length 'L'. When it slips down the plane (without friction) , it takes time 't_1'. When it rolls down the plane with pure rolling , it takes time t_2. The value of $\frac{t_2}{t_1}$ is $\sqrt{\frac{3}{x}}$. The value of x will be ______.

28. A thin rod having a length of 1 m and area of cross section $3\times10^{-6}\ m^2$ is suspended vertically from one end. The rod is cooled form $210°C$ to $160°C$. After cooling , a mass M is attached at the lower end of the rod such that the length of rod again becomes 1 m. Young's modulus and coefficient of linear expansion of the rod are $2\times10^{11}\ Nm^{-2}$ and $2\times10^{-5} K^{-1}$, respectively. The value of M is _________ kg. (Take g = $10\, ms^{-2}$)

29. A uniform heating wire of resistance 36 Ω is connected across a potential difference of 240 V. The wire is then cut into half and potential difference of 240 V is applied across each half separately. The ratio of power dissipation in first case to the total power dissipation in the second case would be 1 : x, where x is ___________.

30. M grams of steam at $100°C$ is mixed with 200 g of ice at its melting point in a thermally insulated container. If it produces liquid water at $40°C$ [heat of vaporization of water is 540 cal/g and heat of fusion of ice is 80 cal/g], the value of M is ____________ grams.

CHEMISTRY MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

31. In which of the following arrangements the order is NOT according to the property indicated against it?

A) $Al^{3+} < Mg^{2+} < Na^{+} < F^{-}$ – increasing ionic size

B) $B < C < N < O$ – increasing first ionization enthalpy

C) $I < Br < F < Cl$ – increasing electron gain enthalpy (with negative sign)

D) $Li < Na < K < Rb$ – increasing metallic radius

32. Assertion : PbI_4 doesnot exist

Reason : $Pb - I$ bond initially formed during the reaction does not release enough energy to unpair $6S^2$ electrons

A) Both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion

B) Both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.

C) The Assertion is correct but Reason is incorrect

D) The Assertion is incorrect and Reason is correct

33. The incorrect order among the following is

A) Bond dissociation enthalpy (kj/mole): $Cl_2 > Br_2 > F_2 > I_2$

B)Melting point /K : $I_2 > Br_2 > Cl_2 > F_2$

C) Melting point/K : $HF > HI > HBr > HCl$

D) Electron gain enthalpy (kj/mole): $Ne > Ar \approx Kr > Xe > He$

34. Which of the following is not arranged in correct sequence?

A) MO, M_2O_3, MO_2, M_2O_5 – decreasing order of basic nature (M = d-block metal)

B) Sc, Ti, V, Cr, Mn – increasing order of highest possible oxidation state

C) d^5, d^3, d^1, d^4 – increasing magnetic moment

D) $Mn^{+2}, Fe^{+2}, Cr^{+2}, Co^{+2}$ – decreasing stability

35. Which of the following is incorrectly matched? (Accoring to CFT)

Complex	Number of unpaired electrons
I) $[FeF_6]^{3-}$	5
II) $[Cr(en)_3]^{2+}$	2
III) $[Co(NH_3)_6]^{3+}$	4
IV) $[Mn(H_2O)_6]^{2+}$	5

A) III B) I C) IV D) II

36. The correct order of dipole moment is:

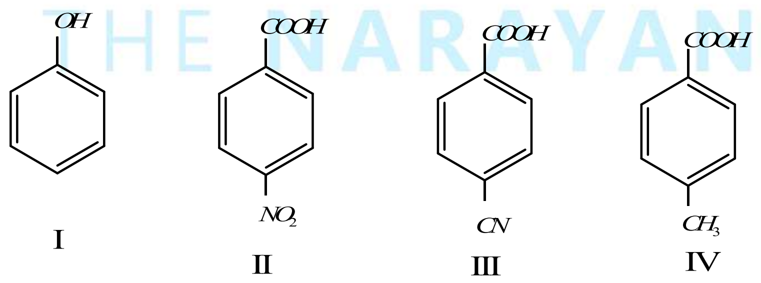
A) $NH_3 < NF_3 < CH_4 < H_2O$

B) $CH_4 < NH_3 < NF_3 < H_2O$

C) $H_2O < NF_3 < NH_3 < CH_4$

D) $CH_4 < NF_3 < NH_3 < H_2O$

37. The correct order of acidic strength character of the following compounds is



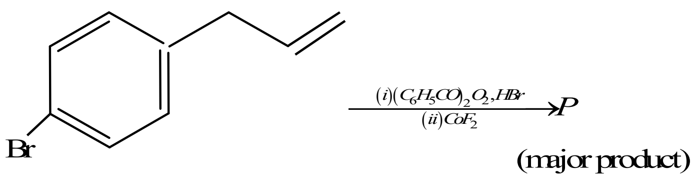
A) IV > III > II > I

B) III > II > I > IV

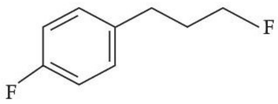
C) II > III > IV> I

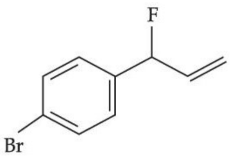
D) I > II > III > IV

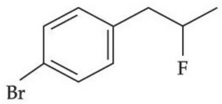
38.

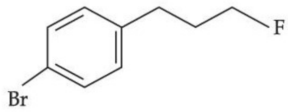


Major product 'P' of above reaction, is:

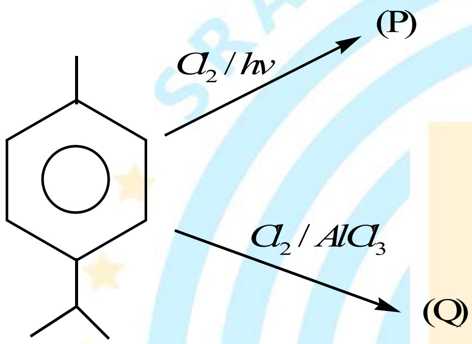
A) 

B) 

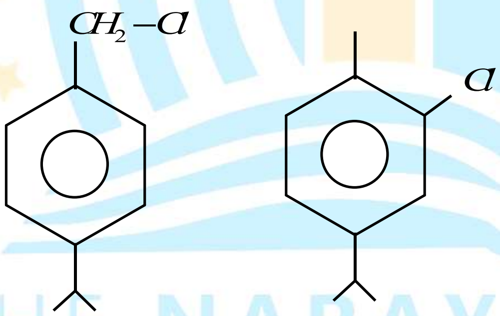
C) 

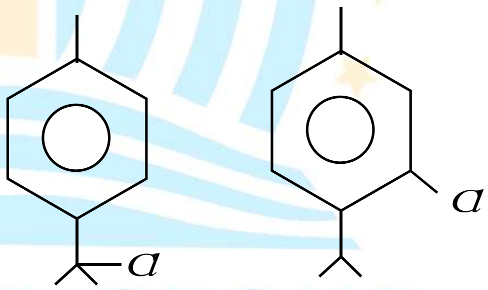
D) 

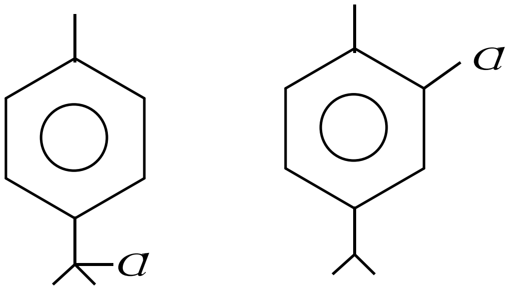
39.

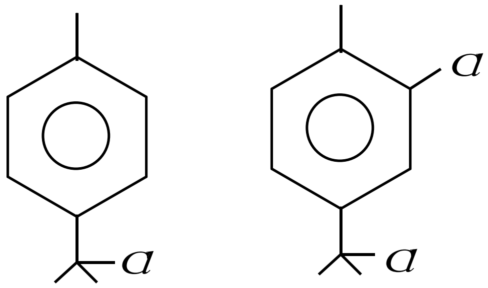


Identify major product of both reactions P and Q respectively

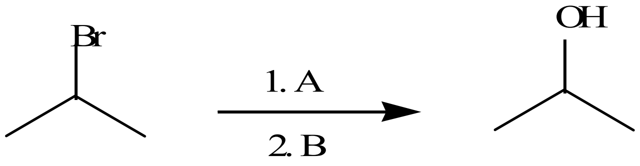
A) 

B) 

C) 

D) 

40. Which of the reagents shown below would accomplish the following transformations?



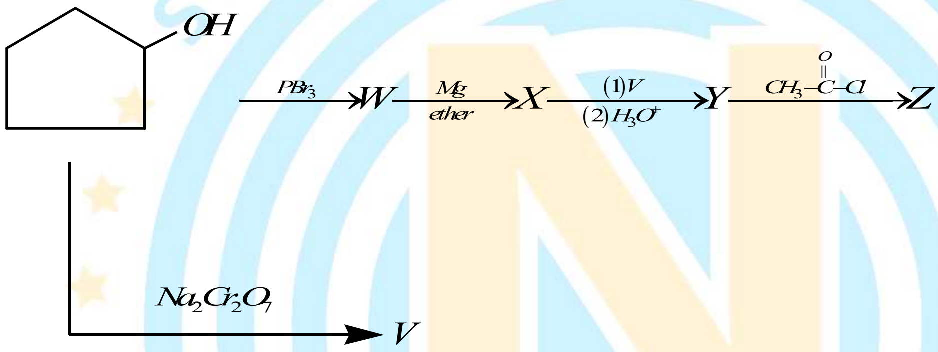
a) A is H_3O^+ ; B is $BH_3 - THF; H_2O_2 / NaOH$

b) A is $NaOH$; B is $BH_3 - THF; H_2O_2 / NaOH$

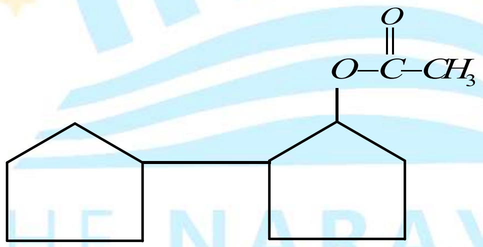
c) A is $HBr\ in\ ether$; B is $Hg(OAC)_2 / H_2O; NaBH_4$

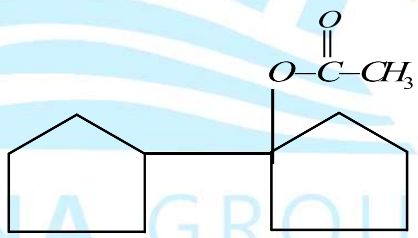
d) A is $NaNH_2$; B is $Hg(OAC)_2 / H_2O; NaBH_4$

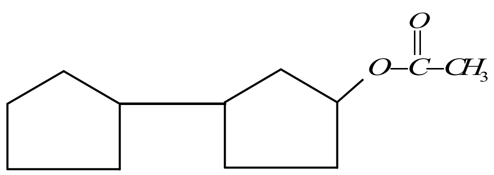
41.

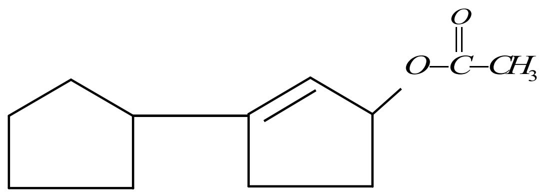


Product Z of above reaction is :

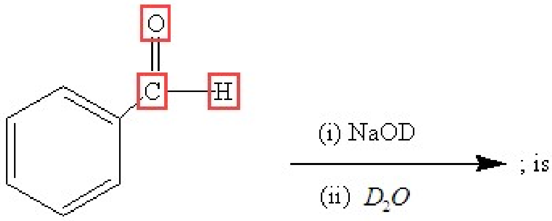
A) 

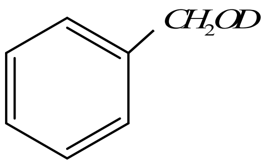
B) 

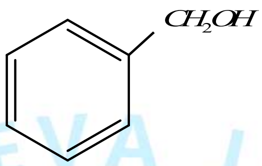
C) 

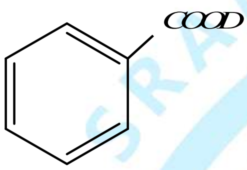
D) 

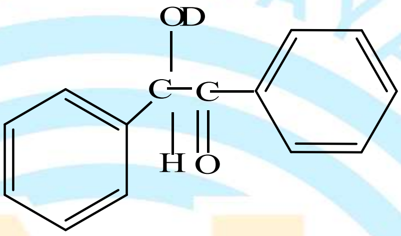
42. The products of the following reaction are,



A) 

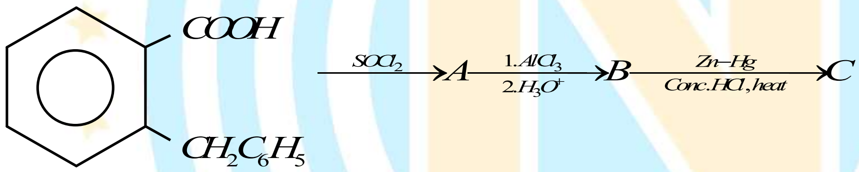
B) 

C) 

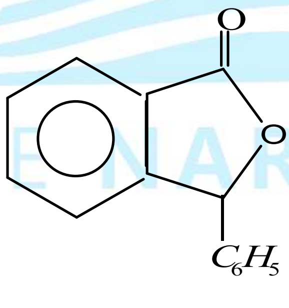
D) 

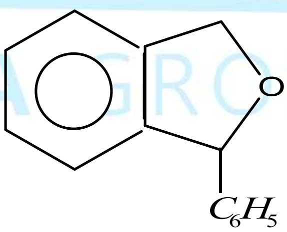
A) A,B B) A,C C) A,D D) C,B

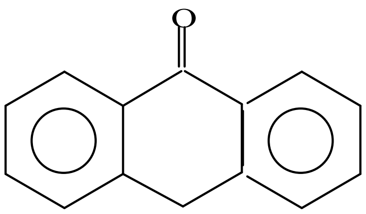
43. Consider the following sequence of reactions,

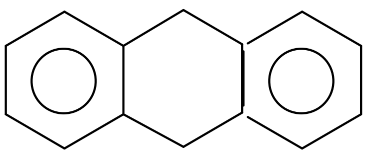


The end product (C) is :

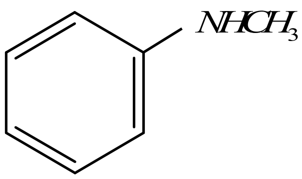
A) 

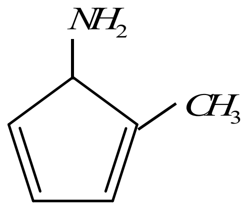
B) 

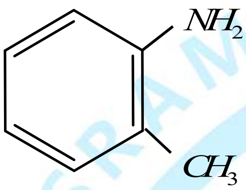
C) 

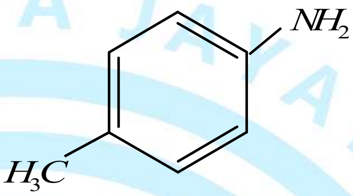
D) 

44. The compound $X(C_7H_9N)$ reacts with benzene sulphonyl chloride to give $y(C_{13}H_{13}NO_2S)$ which is insoluble in alkali. The compound X is :

A) 

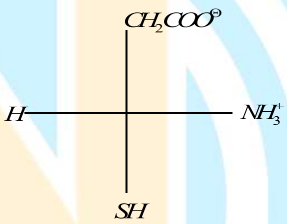
B) 

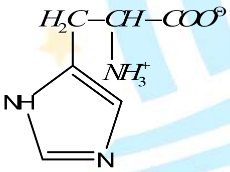
C) 

D) 

45. The amino acid that cannot be obtained by hydrolysis of proteins is-

A) $HOOCCH_2CH\left(\overset{+}{N}H_3\right)COO^-$

B) 

C) 

D) $\overset{+}{N}H_3(CH_2)_4CH(NH_2)COO^-$

46. A certain amount of a reducing agent reduces x mole of $KMnO_4$ and y mole of $K_2Cr_2O_7$ in different experiments in acidic medium. If the change in oxidation state in reducing agent is same in both experiments, $x:y$ is

A) 5:3 B) 3:5 C) 5:6 D) 6:5

47. The ionisation enthalpy of H atoms is $1.312\times10^6 J/mol$ the energy required to excite the electrons in the one of mole of hydrogen atoms from $n=1$ to $n=2$ is

A) $9.84\times10^5 J/mol$

B) $8.51\times10^5 J/mol$

C) $6.56\times10^5 J/mol$

D) $7.56\times10^5 J/mol$

48. $\frac{1}{2}x_2+\frac{3}{2}y_2 \rightarrow xy_3; \Delta H_r = -30kJ$

Entropies of x_2, y_2 *and* xy_3 are 60,40 and 50J $K^{-1}mol^{-1}$ respectively for the reaction. The temperature in kelvin at which the above reaction attains equilibrium.

A)750 B) 650 C) 550 D) 880

49. Match the following if the molecular masses of X,Y,Z and W are same

[Solutions]

Column-I	Column-II
(Solvent and its boiling point)	(Molal elevation constant, K_b)
A) X $(100^0 C)$	P) 0.68
B) Y $(27^0 C)$	Q) 0.53
C) Z $(253^0 C)$	R) 0.98
D) W $(182^0 C)$	S) 0.79

A) $A\rightarrow P; B\rightarrow Q; C\rightarrow R; D\rightarrow S$ B) $A\rightarrow Q; B\rightarrow P; C\rightarrow R; D\rightarrow S$

C) $A\rightarrow P; B\rightarrow Q; C\rightarrow S; D\rightarrow R$ D) $A\rightarrow S; B\rightarrow Q; C\rightarrow R; D\rightarrow P$

50. A solution is 0.1M in Cl^- and 0.001M in CrO_4^{2-}. Solid $AgNO_3$ is gradually added to it. Assuming that the addition does not change in volume and $K_{sp}(AgCl)=1.7\times10^{-10}M^2$ and $K_{sp}(Ag_2CrO_4)=1.9\times10^{-12}M^3$. Select correct statement from the following

A) Ag_2CrO_4 precipitates first because the amount of Ag^+ needed is low.

B) $AgCl$ precipitates first because its K_{sp} is high

C) $AgCl$ will precipitate first as the amount of Ag^+ needed to precipitate is low.

D) Ag_2CrO_4 precipitates first as its K_{sp} is low.

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

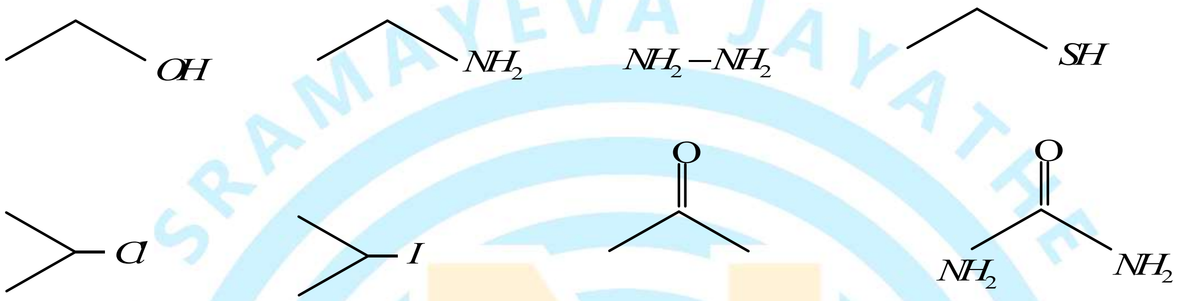
This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Mark nearest Integer only.** **Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

51. Percentage of aniline hydrochloride hydrolysed in its M/40 solution at 25^0C is

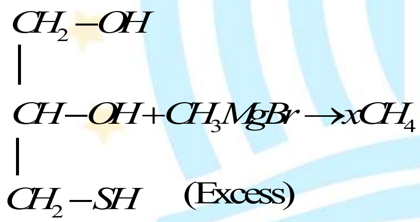
Given ; $\left(E_{C_6H_5NH_2.HCl/H_2} = -0.18V; \frac{2.303RT}{F} = 0.06\right)$

52. How many compounds will give positive Lassaigne's test ?



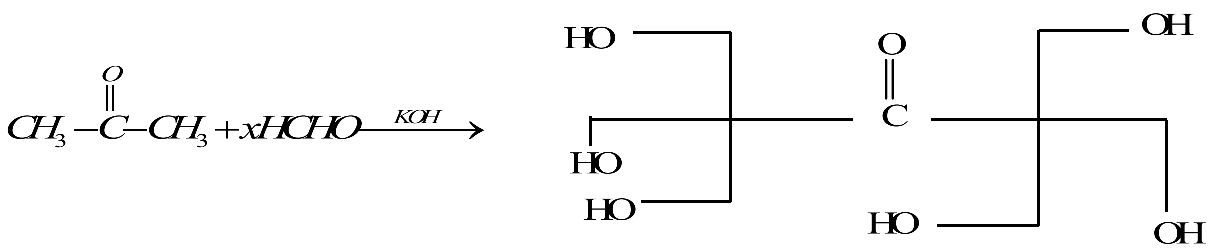
53. ----------- is the magnetic moment of a di valent ion in aqueous solutionswith atomic number 26. (nearest integer)

54.



What is the value of $'x'$in the above reaction?

55.



$x =$ moles of $HCHO$ consumed .

Value of (x)will be

56. a) Glycine b) Valine c) Leucine d) Cysteine

e) Arginine f) Methionine g)Lysine h) Try Ptophan

Sulphur containing amino acid $= x$

Basic amino acid $= y$

Essential amino acid $= Z$

Find the value of $M = \frac{x+y+z}{2}$

57. $NaClO_3$ is used, even in spacecrafts, to produce O_2. The daily consumption of pure O_2 by a person is 492 L at 1atm, 300K. How much amount of $NaClO_3$, in grams, is required to produce O_2 for the daily consumption of a person at 1 atm, 300K? $[Na = 23 gm/mol, Cl = 35.5 gm/mol]$ According to the following reaction.

$$NaClO_3(s) + Fe(s) \rightarrow O_2(g) + NaCl(s) + FeO(s)$$

$R = 0.082 L\ atm\ mol^{-1} K^{-1}$

58. Consider the following reaction

$Na_3PO_4 + (NH_4)_2 MoO_4 + HNO_3(dil) \rightarrow 'X'[canary\ yellow\ PPT]$ then calculate total number of atoms of 15th group elements which are SP^3 hybridized in compound 'X'.

59. How many grams of sucrose (mol. Wt. = 342) should be dissolved up to the nearest integer in 100 gm water in order to produce a solution with 105^0C difference between the freezing point & boiling point temperature at 1 atm?

$(Unit : k_f = 2 K.kg\ mol^{-1}; k_b = 0.5 K.kg\ mol^{-1})$

60. At a certain temperature equilibrium constant (K_c) is 16 for the reaction.

$SO_{2(g)} + NO_{2(g)} \Leftrightarrow SO_{3(g)} + NO_{(g)}$. If we take one mole each of all the four gases in one litre container, then the % moles of NO_2 at equilibrium is

MATHEMATICS

MAX.MARKS: 100

SECTION – I

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

61. If z is a complex number satisfying the equation $|z+i|+|z-i| = 8$, on the complex plane then maximum value of $|z|$ is

A) 2 B) 4 C) 6 D) 8

62. The system of equations

px+(p+1)y+(p-1)z = 0

(p+1)x+py+(p+2)z=0

(p-1)x+(p+2)y+pz=0 has a non – trivial solutions for

A) Exactly two real values of p. B) no real value of p.

C) Exactly one real value of p. D) infinitely many real value of P.

63. If two whole numbers are randomly selected and multiplied then the chance that the unit place in their product is 0 or 5, is

A) 20% B) 16% C) 61% D) 36%

64. Consider the word 'HALEAKALA'. The number of ways the letters of this word can be arranged if

	Column-I		Column-II
a)	All 'A' are separated	p)	5
b)	Keeping the position of each consonant fixed	q)	60
c)	All the vowels occur together	r)	300
d)	No two vowels are consecutive	s)	900

Code:

A) $a-s, b-p, c-p, d-r$ B) $a-q, b-r, c-s, d-p$

C) $a-s, b-p, c-r, d-q$ D) $a-q, b-r, c-s, d-s$

65. **Statement-I:** Out of 21 tickets with numbers 1 to 21,3 tickets are drawn at random, the chance that the numbers on them are in A.P is $\frac{10}{133}$.

Statement-II: Out of $(2n+1)$ tickets consecutively numbered, three are drawn at random, the chance that the numbers on them are in A.P is $(4n-10)/(4n^2-1)$.

A) Statement- I is true, Statement-II is true, Statement-II is correct explanation of for statement-I

B) Statement- I is true, Statement-II is true, Statement-II is not correct explanation of for statement-I

C) Statement- I is true, Statement-II is false

D) Statement- I is false, Statement-II is true

66. A cubic polynomial y = f(x) is such that A(-1, 3) and B (1, -1) are the relative a maximum and relative minimum points respectively. The value of f(2) is

A) 6 B) 4 C) 0 D) 3

67. If the surface area of a sphere of radius r is increasing uniformly at the rate 8cm^2/sec, then the rate of change of its volume is

A) Proportional to $\sqrt{r}$ B) Constant

C) Proportional to r D) Proportional to r^2

68. If $\vec{V_1} = \hat{i} + \hat{j} + \hat{k}$; $\vec{V_2} = a\hat{i} + b\hat{j} + c\hat{k}$ where a, b, c $\in$ {-2, -1, 0, 1, 2}, then find the number of non-zero vectors $\vec{V_2}$ which are perpendicular to $\vec{V_1}$

A) 18 B)24 C) 21 D) 12

69. Let L be the line passing through the point P (1,2) such that its intercepted segment between the Co-ordinate axes is bisected at P. If L_1 is the line perpendicular to L and passing through the point $(-2,1)$ then the point of intersection of L and L_1 is

A) $\left(\frac{3}{5}, \frac{23}{10}\right)$ B) $\left(\frac{4}{5}, \frac{12}{5}\right)$ C) $\left(\frac{11}{20}, \frac{29}{10}\right)$ D) $\left(\frac{3}{10}, \frac{17}{5}\right)$

70. Let $f:R\to R$ be defined by $f(x)=x^3+3x+1$ and g is the inverse of 'f' then the value of $g''(5)$ is equal to

A) $\frac{-1}{6}$ B) $\frac{-1}{36}$ C) $\frac{1}{6}$ D) $\frac{1}{36}$

71. If $\int\left(x^{24}+x^{16}+x^8\right)\left(2x^{16}+3x^8+6\right)^{1/8}dx=\frac{1}{\alpha}\left(2x^{24}+3x^{16}+6x^8\right)^{\frac{\beta}{\gamma}}+C$ (Where C is constant of integration and β, γ are coprime numbers) then the value of $(\alpha+\beta+\gamma)$ is…

A) 51 B) 61 C) 71 D) 81

72. Let y = y(x) satisfy the equation $y(x)+\int_1^x y(t)dt=x^2$. The value of y(e) is equal to

A) $2e-2+e^{1-e}$ B) $2e-1+e^{1-e}$ C) $2e+2+e^{1-e}$ D) $2e+1+e^{1-e}$

73. If the roots of the equation $72x^3-108x^2+46x-5=0$ are in A.P, then the difference between largest and smallest root lies in the interval

A) $\left(0,\frac{1}{2}\right)$ B) $\left(\frac{1}{2},\frac{1}{\sqrt{2}}\right)$ C) $\left(\frac{1}{\sqrt{2}},1\right)$ D) $(1,2)$

74. If the mean deviation about the median of the numbers a, 2a,…… 50a is 50, then $|a|$ equals:

A) 1 B) 2 C) 3 D) 4

75. The value of $\sum_{k=1}^{13}\frac{1}{\sin\left(\frac{\pi}{4}+\frac{(k-1)\pi}{6}\right).\sin\left(\frac{\pi}{4}+\frac{k\pi}{6}\right)}$ is equal to

A) $3-\sqrt{3}$ B) $2\left(3-\sqrt{3}\right)$ C) $2\left(\sqrt{3}-1\right)$ D) $2\left(2+\sqrt{3}\right)$

76. Suppose $A_1, A_2, \ldots, A_{30}$ are thirty sets each having 5 elements and $B_1, B_2, \ldots, B_n$ are n sets each with 3 elements, let $\bigcup_{i=1}^{30}A_i=\bigcup_{i=1}^{n}B_j=S$ and each elements of S belongs to exactly 10 of the $A_i's$ and exactly 9 of the $B_j's$. Then, n is equal to

A) 40 B) 54 C) 45 D) 27

77. Let A be a non- singular matrix such that $A^{-1}=\begin{pmatrix}1 & 1 & 1\\ 4 & 6 & 8\\ 100 & 100 & 99\end{pmatrix}$, then the value of $\det\left(Adj(2A)\right)+\det(2A)$ is

A) 8 B) 12 C) 16 D) 28

78. Let $\alpha,\beta(\alpha>\beta)$ be the roots of the quadratic equation $x^2-x-4=0$. If $P_n=\alpha^n-\beta^n$. $n\in N$ then $\frac{P_{15}P_{16}-P_{14}P_{16}-P_{15}^2+P_{14}P_{15}}{P_{13}P_{14}}$ is equal to

A) 15 B) 16 C) 17 D) 18

79. Let $f(x)=[x]+|1-x|$ for $-1\le x\le 3$ where [x] is the integral prat of x. Then the number of value of x in [-1,3] at which f is not continuous is .

A) 0 B) 1 C) 2 D) 4

80. $\cos^{-1}\left(\cos(-5)\right)+\sin^{-1}\left(\sin(6)\right)-\tan^{-1}\left(\tan(12)\right)$ is equal to :

(The inverse trigonometric functions take the principal values)

A) $3\pi-11$ B) $4\pi-9$ C) $4\pi-11$ D) $3\pi+1$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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Marking scheme: +4 for correct answer, -1 in all other cases.

81. Let $\omega=e^{\frac{2i\pi}{3}}$ and a, b, c, x, y, z be non – zero complex number such that

a + b + c = x

a + bω + cω^2=y

a + bω^2+cω=z

Then the value of $\frac{|x|^2+|y|^2+|z|^2}{|a|^2+|b|^2+|c|^2}$, is equal to

82. If $f(x)=ax^3-9x^2+9x+3$ is strictly increasing on R, then find the number of integralvalues of $'a'$, $a\in[-5,100]$

83. The shortest distance between the lines x = y = z and $\frac{x-0}{1}=\frac{y-1}{-1}=\frac{z}{0}$ is $\frac{1}{\sqrt{\lambda}}$ then $\lambda=$

84. The locus of image of the point $(\lambda^2,2\lambda)$ in the line mirror x–y+1= 0 where λ parameter is $(x-a)^2=b(y-c)$ where a, b, c $\in$I, then find the value of (a + b + c).

85. Two vertices of an equilateral triangle are (-1,0) and (1,0), and its third vertex lies above the x – axis. If the equation of its circumcircle is $x^2+y^2-\frac{2y}{\sqrt{3}}-\lambda=0$, then find the value of λ.

86. $\int_0^1(1-x^7)^{\frac{1}{4}}dx-\int_0^1(1-x^4)^{\frac{1}{7}}dx$ is equal to

87. Find the area of the region bounded by $y=(x-4)^2$, $y=16-x^2$ and the $x-$axis.

88. Let a and b are two positive real numbers such that $a^2 + b = 2$, then the maximum value of term independent of x in the expansion of $(ax^{1/6} + bx^{-1/3})^9$ is

89. If $\lim_{x\to\infty}x^{\alpha}\left(\sqrt{x^2+\sqrt{x^4+1}}-\sqrt{2}x\right)$ exist and has value non – zero finite number L, then find the value of $\frac{\alpha}{L^2}$.

90. Let $A=\{1,2,3,4\}$ and $B=\{1,2,3,4\}$. If $f:A\to B$ is an one – one function and $f(x)\neq x$ for all $x\in A$, then the number of such possible functions, is